THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A car and a truck, both with the same momentum, are brought to rest in the same time. The retarding force is: 1

(a) the same for both **(b)** greater for the truck
(c) greater for the car **(d)** zero

2. A body of mass 4 kg moving at 5 m/s is brought to rest by a force in 2 s. The force is: 1

(a) 40 N **(b)** 20 N **(c)** 10 N **(d)** 2.5 N

3. Assertion (A): The mass of a body is the same on the Earth and on the Moon. Reason (R): Mass depends on the gravitational pull at the place. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A horse of mass 400 kg pulls a cart of mass 200 kg with an acceleration of 1 m/s squared; the friction on the cart is 100 N. The pull of the horse on the cart is: 1

(a) 100 N **(b)** 200 N **(c)** 300 N **(d)** 600 N

5. Two bodies of masses 2 kg and 1 kg are pushed by equal forces for the same time from rest. The ratio of their final velocities (2 kg : 1 kg) is: 1

(a) 4 : 1 **(b)** 1 : 2 **(c)** 2 : 1 **(d)** 1 : 1

6. A force of 20 N acts on a body for 0.5 s. The impulse is: 1

(a) 0.5 N s **(b)** 10 N s **(c)** 20 N s **(d)** 40 N s

7. Assertion (A): A body moving with constant velocity has no net force on it. Reason (R): Zero acceleration implies zero net force. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A 2 kg block and a 3 kg block are in contact on a smooth floor and pushed by 10 N on the first. The force between them is: 1

(a) 4 N **(b)** 10 N **(c)** 6 N **(d)** 5 N

9. **Assertion (A):** The mass of a body is the same everywhere, whereas its weight changes from place to place. 1

Reason (R): The weight of a body is independent of the acceleration due to gravity.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Action and reaction forces cancel each other because they are equal and opposite. 1
Reason (R): Action and reaction forces act on different bodies.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A force of 40 N acts on a body of mass 8 kg. Find the acceleration. 2

12. A boy kicks a football of mass 0.5 kg with 10 N for 0.1 s from rest. Find its speed. 2

13. A rocket expels gas at 2000 m/s at the rate of 5 kg/s. Find the thrust. 2

14. Why is it dangerous to jump from a moving train? Use inertia. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A force of 60 N acts on a body of mass 12 kg. Find the acceleration and the velocity gained in 3 s from rest. 3

16. A 50 kg box needs a force of 250 N to just start moving. When it moves, a pull of 300 N gives it a constant acceleration against a friction of 200 N. Find the static friction limit and the acceleration. 3

17. A 20 kg block slides down a smooth slope with a force of 60 N along the slope. Find the acceleration and the speed after 5 s. 3

OR

A force acts on a 2 kg body at rest as shown by the graph: 10 N for 2 s and then 0. Find the speed after 2 s, the speed after 5 s and the distance in the first 5 s. 3

18. Explain how the recoil velocity depends on the mass of the gun with the example of two guns of 2 kg and 8 kg firing the same bullet. 3

OR

A 3 kg body moves on a smooth floor with a speed of 4 m/s. A force of 6 N then acts on it in the direction of motion for 2 s. Find the new speed and the change in momentum. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A cricket ball of mass 0.15 kg moving at 30 m/s is caught by a fielder in Chennai. If he stops the ball in 0.05 s with stiff hands, the hands sting. A coach advises him to pull his hands back while catching, which makes the stopping time 0.15 s. 5
(i) Find the momentum of the ball before it is caught.
(ii) Find the average force on the hands if the ball is stopped in 0.05 s.
(iii) Find the force if the stopping time is 0.15 s.
(iv) Name the law used and explain why pulling the hands back reduces the force.

20. A car of mass 900 kg is brought to rest from 15 m/s in a collision lasting 0.3 s with and without airbags 0.1 s. Find the average forces on a 60 kg passenger in the two cases and compare. 5

OR

Discuss the application of Newton's laws to the motion of a car: engine force, friction, air resistance, inertia in braking, and the safety measures including seat belts, airbags, ABS and speed limits. 5

END OF QUESTION PAPER — Total: 40 Marks